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Dynamic phase transition in 1T-TaS_2 via a thermal quench

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Ultrafast light–matter interaction has emerged as a mechanism to control the macroscopic properties of quantum materials. However, technological applications of photoinduced phases are limited by their ultrashort lifetimes and the low temperatures required for their stabilization. One such phase is the hidden metallic charge density wave state in 1T-TaS_2, whose origin and stability above cryogenic temperatures remain the subject of debate. Here, we demonstrate that this phase can be stabilized at thermal equilibrium by accessing a mixed charge density wave order regime through thermal quenching. Using X-ray high-dynamic-range reciprocal space mapping and scanning tunnelling spectroscopy, we reveal the coexistence of commensurate charge density wave and hidden metallic charge density wave domains up to 210 K. Our findings show that each order parameter breaks basal plane mirror symmetry with different chiral orientations and induces out-of-plane unit cell tripling in the hidden phase. Despite metallic domain walls and a finite density of states, the bulk resistance remains insulating due to charge density wave stacking disorder. Our results establish the hidden state as a thermally stable phase and introduce an alternative mechanism for switchable metallic behaviour in thin flakes of 1T-TaS_2 and similar materials with competing phases.

The correlated van der Waals material 1T-TaS_2 displays a series of first-order phase transitions associated with the onset of charge density wave (CDW) orders with different degrees of commensurability[1–3]. On reducing the temperature, at $T = 550$ K, a mirror-symmetry-preserving incommensurate (I) CDW develops from a metallic state[4]. At $T_{\text{I-NC}} \approx 350$ K, 1T-TaS_2 enters into a nearly commensurate (NC) CDW (NC-CDW) phase. The atomic rearrangement associated with the incommensurate NC-CDW (in-plane ordering wavevector $q_{||} = 0.284$ reciprocal lattice units (r.l.u.)) leads to disconnected regions displaying a $\sqrt{13} \times \sqrt{13}$ spatial reconstruction in which 12 Ta atoms move towards a central one to form a star-of-David pattern[5]. The NC-CDW breaks the mirror symmetries of the lattice basal plane and spontaneously selects one of two chiral domains rotated from the a axis by $\phi \approx 12°$. Then, at $T_{\text{NC-C}} \approx 175$ K when cooling or $T_{\text{NC-C}} \approx 225$ K when warming, a three-dimensional CDW order emerges from the NC-CDW phase. This low-temperature phase is commensurate (C) with the atomic lattice in-plane ($\phi = 13.9°$ and $q_{||} = 0.277$ r.l.u.) with the in-plane C-CDW atomic reconstruction inheriting the chiral structure from the NC-CDW. On the other hand, in the out-of-plane direction, the C-CDW displays a large stacking disorder between dimerized CDW planes[1,2].

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Experimental probes spanning a large range of length scales have found that in the C-CDW phase of 1T-TaS_2, samples lack the chiral twin domains expected across the $T_{NC\text{-}C}$ first-order phase transition[6–10]. The stabilization of a single chiral domain reflects the kinetics of cooling across the only first-order phase transition in 1T-TaS_2 that breaks in-plane mirror symmetries, $T_{I\text{-}NC}$ (refs. 11–13). During synthesis, the metastable 1T phase is stabilized by quenching from $T \approx 1{,}000$ to 273 K across the polytype transition, $T_{1T\text{-}2H} = 600$ K (ref. 1). During this initial quench, the cooling rate across $T_{I\text{-}NC}$ is effectively slow when compared to the intrinsic timescales of the electronic (femtoseconds) and lattice (picoseconds) degrees of freedom and enables the nucleation of large chiral CDW domains with lateral sizes comparable to the correlation length of the order parameter[14]. The chiral mono-domain structure of the C-CDW in 1T-TaS_2 is stable to cooling through $T_{NC\text{-}C}$ (refs. 6–10), with suggestions that strain fields and defects can pin a single chirality[6,15]. Still, chirality reversal of single-domain samples has been demonstrated by repeated cleaving of a bulk sample[7] or by cycling through $T_{I\text{-}NC}$ (ref. 16).

On the other hand, multi-chiral domain samples can be engineered in thin exfoliated flakes by heating and quenching above $T_{1T\text{-}2H}$ (refs. 17,18) or upon irradiation with 800-nm laser pulses with fluence $F > 7$ mJ cm^{-2} (refs. 6,19). Once created, chiral domain walls are long-lived at low temperatures, but a dominant chiral domain of dimensions comparable to the sample size can be restored by heating above $T_{I\text{-}C}$ and adiabatically cooling through the mirror-symmetry-breaking IC–NC transition[6,17,18]. Thus, $T_{I\text{-}NC}$ emerges as the relevant energy scale for chiral CDW domain nucleation in 1T-TaS_2 as the temperature at which the initial mirror-symmetry-breaking transition in the CDW phase diagram[1] occurs and the largest specific heat divergence is observed[20]. Moreover, X-ray pair distribution function measurements have revealed that the anisotropic Ta Debye–Waller factor, associated with CDW formation, remains constant across the NC–C CDW transition but dramatically changes across $T_{I\text{-}NC}$ (ref. 21).

Another low-temperature metastable metallic CDW phase, the so-called hidden H-CDW phase, has been accessed in 1T-TaS_2 from the insulating C-CDW state with a single femtosecond laser pulse with a fluence larger than a critical value ($F_c > 1$ mJ cm^{-2})[22–24] or by exciting 1T-TaS_2 with a nanosecond voltage pulse[25–29]. The H-CDW phase locally resembles the star-of-David arrangement but breaks the large C-CDW domains into many nanoscale domains and has a metallic density of states[26,30,31]. These domains maintain the chirality of the C-CDW[23] and are related by a relative translation of the star-of-David pattern by a finite number of atomic lattice vectors[26,30,32]. Slight differences between the displacement of the Ta atoms between the H-CDW and the C-CDW distinguish the two. The H-CDW phases result in CDW Bragg peaks rotated by $\phi = 11.9°$ from the a^* axis with a larger basal plane ordering wavevector, $q_{||} = 0.284$ r.l.u. The CDW plane stacking also rearranges along the l-direction with an $l = 1/3$ periodicity[23]. Note that a similar metallic mosaic state can also be induced by cooling thin exfoliated samples (thickness $d < 24$ nm) across $T_{I\text{-}NC}$ with a cooling rate faster than 0.2 K s^{-1} (ref. 33) or by intrinsic strain and defect effects[15,34,35].

Despite much study, the origin of the H-CDW state remains controversial, with photodoping[30,36], incoherent thermal effects[26,37] or electron–phonon relaxation bottlenecks[31,38] all being possible explanations. Nonetheless, the H-CDW has attracted much interest, not only as an example of the non-thermal control of quantum materials[39] but also as a platform for new functional devices[40–42]. However, the H-CDW has been stabilized only for ultrashort periods (~100 μs) at temperatures above $T = 50$ K (refs. 22,31,43), preventing its deployment in efficient and fast memory devices for new technological applications[44].

Here, we demonstrate that the H-CDW phase exists in thermal equilibrium in 1T-TaS_2. We have extended its thermal stability by a factor of three by accessing a new long-lived mixed-CDW state with a reversible thermal quench across $T_{I\text{-}NC}$. We used X-ray high-dynamic-range reciprocal space mapping (HDRM) to unequivocally demonstrate the coexistence of the C-CDW and H-CDW states in thermally quenched samples. Each CDW order parameter is linked to a mirror-symmetry-breaking chiral domain. The mixed-CDW state emerges from a multi-domain NC-CDW state below $T^{CD} = 180$ K during cooling and $T^{WU} = 210$ K during warming. The mixed-CDW low-temperature state can be recovered if the system is not heated above $T_{I\text{-}NC}$. Scanning tunnelling spectroscopy (STS) shows that the mixed-CDW state is characterized by a semimetallic in-plane density of states. At the same time, the bulk resistivity remains insulating due to out-of-plane CDW stacking disorder in mixed-CDW bulk samples. Our results and calculations of the free-energy landscape within Ginzburg–Landau theory demonstrate a new thermally accessible state in the phase diagram of 1T-TaS_2, which may enable a new mechanism to engineer a semimetallic density of states at temperatures well above liquid nitrogen (LN_2) temperatures.

Results

Quenched transition reveals new mixed-CDW phase

Figure 1a shows a characteristic reciprocal space map for the $[H, K, 0]$ plane for the C-CDW phase of 1T-TaS_2, collected after a standard cooling cycle. Red circles highlight the CDW reflections at the ordering vectors $\mathbf{q}_1^C = (\sigma_1^C, \sigma_2^C, l)$ and $\mathbf{q}_2^C = (-\sigma_2^C, \sigma_1^C + \sigma_2^C +, l)$ with $\sigma_1^C = 3/13$ r.l.u. and $\sigma_2^C = 1/13$ r.l.u. (refs. 1,2), corresponding to a single C-CDW chiral domain. Other non-indexable reflections in the reciprocal space map are attributed to a second small crystallite in our sample. Identical reciprocal space maps were obtained across many cooling cycles and samples. We found that the C-CDW domain structure in a given sample depends on the cooling rate through $T_{I\text{-}NC}$. A fast thermal quench across $T_{I\text{-}NC}$ stabilizes a multi-chiral domain CDW. Figure 1b shows the $[H, K, 0]$ reciprocal space map after an in situ quench in our bulk sample across $T_{I\text{-}NC}$ down to $T = 140$ K at a rate of 0.2 K ms^{-1} near $T_{I\text{-}NC}$ (see Methods for a description of the in situ quench method and the calculation of quenching rates using heat diffusion calculations). We found that thermal quenching consistently resulted in a new set of superlattice reflections of opposite chirality to the original set of satellite peaks ($\phi = -12.3°$) (Fig. 1b, highlighted by blue markers). The superior reciprocal space resolution of synchrotron single-crystal HDRM enabled us to resolve differences in the wavevector of the two sets of CDW Bragg peaks with different chiralities. The in-plane wavevector magnitude $|q_{||}| = 0.284(3)$ Å^{-1} of the quench-induced satellite peaks was larger than that of the C-CDW ($|q_{||}| = 0.277(3)$ Å^{-1}) and was the same as the light-stabilized H-CDW state[23]. This is shown in Fig. 1c, which displays the integrated intensity across the C-CDW and H-CDW Bragg peaks (dashed lines in Fig. 1a,b) as a function of $|q_{||}|$. Strikingly, each order parameter preferentially selected a chirality orientation, with the C-CDW domains of the mixed-CDW phase maintaining that of the as-grown samples.

We found that the c-axis periodicity differed between the C-CDW and H-CDW-like domains. Figure 1d,e show $[H, 0, L]$ reciprocal space maps centred at $H = 1$ r.l.u. and $L = 0$ r.l.u. ($\delta k = 0.3$ r.l.u.) before and after the thermal quench, respectively. The as-grown sample had a broad two-peak structure, with the centre of mass at integer values of L. These short-range out-of-plane correlations are characteristic of the dimerization of CDW layers in the low-temperature C-CDW phase[45]. On the other hand, although they retained a dumbbell-like shape, the H-CDW-like satellite peaks shifted towards $l = \pm 1/3$, indicating that there was a collapse of the interlayer dimerization towards a triple-layer CDW stacking, as was observed for the light-induced H-CDW[23]. The H-CDW-like satellite peaks are sharper than the C-CDW ones, reflecting a longer correlation length for the quench-induced domains $\xi = 6(6)$ nm (Fig. 1f). A similar sharpening of the out-of-plane CDW peaks has also been observed in light-induced H-CDW 1T-TaS_2 (ref. 23). Altogether, our HDRM data for quenched samples demonstrate the coexistence of the C-CDW with a second CDW phase with the same order parameter as the H-CDW phase at much higher temperatures ($T = 140$ K).

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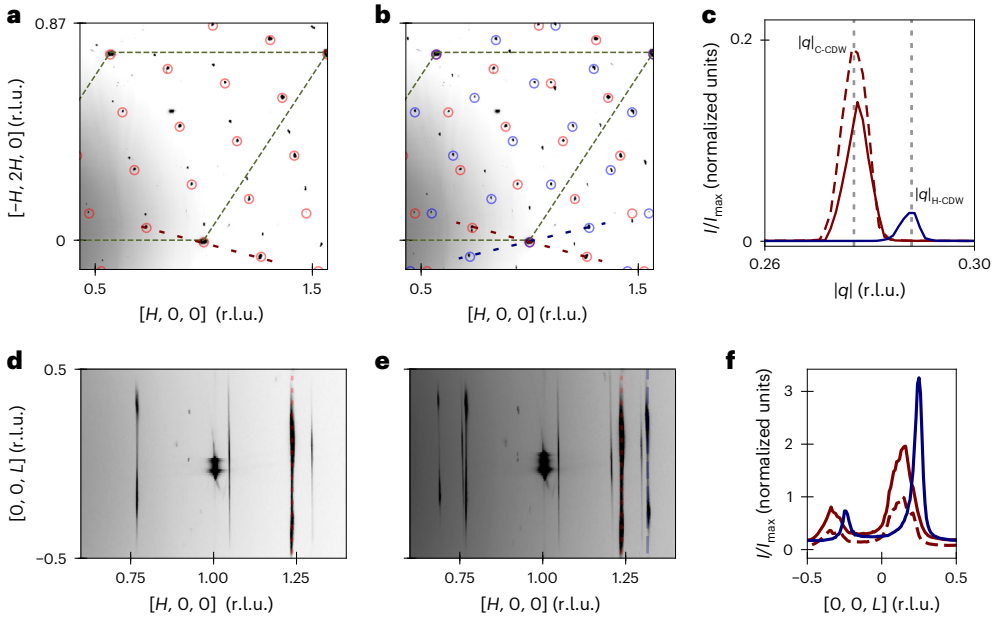

Fig. 1 | Thermal quench into the mixed-CDW phase. a, Reciprocal space map parallel to the [*H*, *K*, 0] plane (δ*L* = 0.5 and *L* = 0) at *T* = 140 K for an as-grown sample. **b**, Reciprocal space map parallel to the [*H*, *K*, 0] plane (δ*L* = 0.5 and *L* = 0) at *T* = 80 K after a quench. Red and blue markers highlight the C-CDW and H-CDW-like satellite peaks, respectively. Green dashed lines highlight the a^* and b^* directions. **c**, Average intensity (*I*) dependence as a function of |*q*| across a C-CDW satellite peak in the as-grown sample (dashed red line) and the C-CDW (red line) and H-CDW-like (blue line) after a quench (red and blue dashed lines in **b**). Vertical dashed lines correspond to the reported $|q_{||}|$ values for the C-CDW[1] and H-CDW[23] satellite peaks. **d**, Reciprocal space map parallel to the [*H*, 0, *L*] plane for the as-grown sample. **e**, Reciprocal space map parallel to the [*H*, 0, *L*] plane for the quenched sample. Vertical red and blue dashed lines indicate C-CDW and H-CDW-like satellite peaks, respectively. **f**, Integrated intensity of a satellite peak along [0, 0, *L*] for the as-grown sample (red dashed line) and after a quench (red line, C-CDW peak and blue line, H-CDW peak).

Critical temperature and stability of the mixed-CDW state

We performed a set of in situ thermal cycles to directly demonstrate that (1) $T_{\text{I-NC}}$ is the relevant temperature scale for accessing the mixed state and (2) the transition is reversible. Consecutive thermal quenches were performed at different quench rates and starting temperatures, as illustrated in Fig. 2a (Methods and Extended Data Fig. 1). We found that after quenching from above $T_{\text{I-NC}}$, the mixed-CDW state was thermally stable and insensitive to warming and cooling cycles if the temperature was kept less than $T_{\text{I-NC}}$ (Fig. 2b). Such thermal stability is in contrast with the light-induced H-CDW phase, which was suppressed above *T* = 60 K (ref. 22). However, the C-CDW state could be recovered from the mixed state by warming up and slow-cooling across $T_{\text{I-NC}}$ (Fig. 2c). Subsequent thermal quenches from *T* = 300 K did not recover the mixed state (Fig. 2d), which could be recovered only upon heating and quenching from above $T_{\text{I-NC}}$ (Fig. 2e). Thus, our data confirm the IC–NC CDW as the necessary energy scale for accessing the mixed-CDW phase containing both the C-CDW and H-CDW order parameters.

The possibility of recovering the H-CDW phase as part of the stable mixed-CDW phase by keeping the samples below $T_{\text{I-NC}}$ enabled us to form the CDW mixed state ex situ (Methods) and to perform further characterization of this state. Note that the mixed-CDW state was stable over months, independent of the number of cleaves, and has been reproduced in quenched samples over several growth batches. Detailed temperature-dependent X-ray diffraction (XRD) measurements (Methods) of ex situ quenched samples display 12 satellite reflections characteristic of the mixed-CDW state at low temperatures (Fig. 2f). For temperatures above $T^{\text{WU}}_{\text{NC-Mixed}} = 210$ K during warming or below $T^{\text{CD}}_{\text{NC-mixed}} = 195$ K during cooling, we observed a transition into the NC-CDW state. However, the NC-CDW state in quenched samples was characterized by two sets of CDW Bragg peaks with opposite chiralities. This state is shown in Fig. 2f at *T* = 270 K for ex situ quenched samples. Equivalent HDRM data at *T* = 300 K are shown in Extended Data Fig. 2. Note that the L dependence of the two NC-CDW satellite peaks is different from that for H-CDW we observe at low temperature (Extended Data Fig. 2), ruling out the presence of a supercooled NC-CDW phase in our quenched samples[33]. The existence of two chiral domains in the NC-CDW is consistent with previous ultrafast electron diffraction experiments that demonstrated that the chirality of the low-temperature C-CDW domains was inherited from the symmetry-breaking at the IC–NC transition[46] and that two NC-CDW domains were stabilized after laser excitation at room temperature[6].

The detailed temperature dependence of the H-CDW-like and NC-CDW Bragg reflections are shown in Fig. 2h,i. Upon stabilizing the mixed state at low temperatures with a quench across $T_{\text{I-NC}}$, the H-CDW peaks directly emerged from NC-CDW peaks of corresponding chirality but with a different out-of-plane periodicity (Fig. 1f and Extended Data Fig. 2), indicating a different CDW out-of-plane stacking order between the two orders. This temperature dependence was distinct from that of the metastable H-CDW phase, which was suppressed above *T* = 60 K (refs. 22,23,30). We found that C-CDW and H-CDW-like Bragg peaks show the same temperature dependence (Extended Data Fig. 3), which indicates the coupled nature of the two order parameters in the mixed-CDW phase. The asymmetrically narrower hysteretic loop of the first-order transition, when compared to as-grown samples, $T^{\text{CD}}_{\text{NC-C}} = 180$ K and $T^{\text{WU}}_{\text{NC-C}} = 210$ K, in conjunction with the narrower CDW Bragg peaks along *L* and the temperature dependence of the [0010] structural Bragg peak shown in Fig. 2c, indicates longer-range interlayer correlations and a more three-dimensional nature for the H-CDW phase.

Scanning tunnelling microscopy (STM), measurements confirmed the multi-domain nature of our quenched samples. Figure 3a shows a characteristic 80 × 80-nm^2 area of a quenched sample. Each domain shows the atomic reconstruction of the sulfur layer due to a low-temperature CDW, as shown in Fig. 3c for *V* = 380 mV and *I* = 0.35 nA with a scale bar of length 2 nm. Note that the in-plane momentum resolution of our STM experiments prevented us from distinguishing

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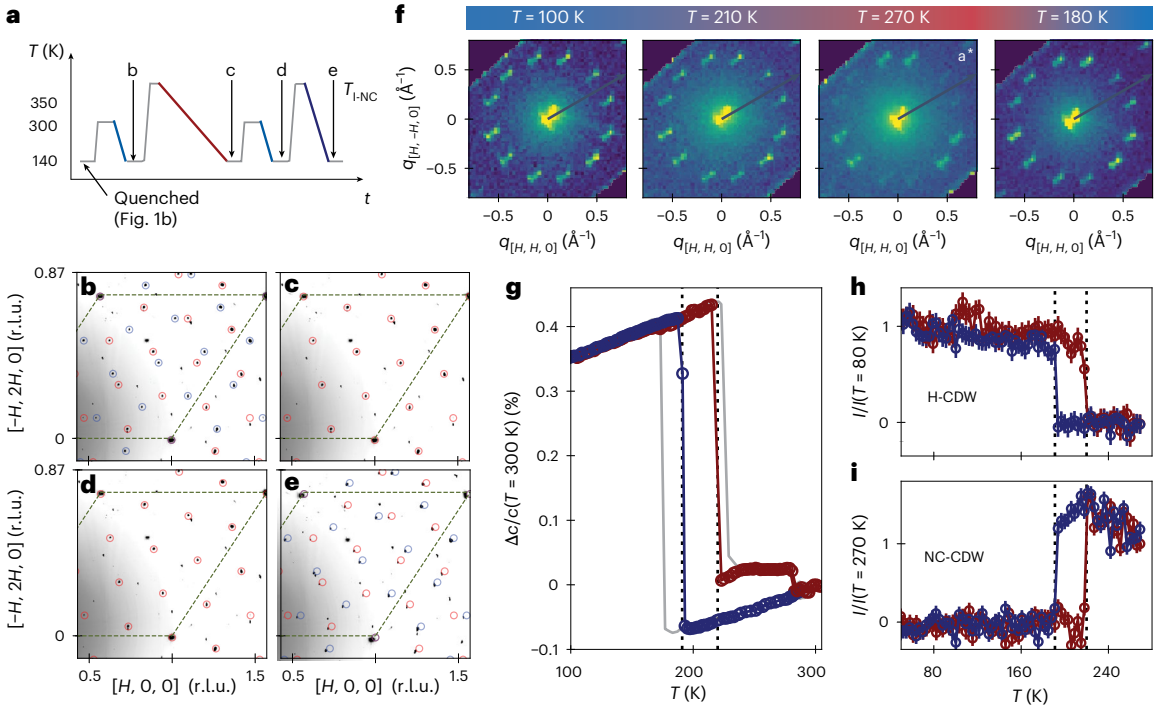

Fig. 2 | Temperature dependence of the mixed-CDW state. a, Schematic of the subsequent in situ thermal cycles undergone by a pre-quenched sample. **b**–**e**, HDRM reciprocal space maps parallel to the [*H*, *K*, 0] plane ($\delta L = 0.5$ and $L = 0$) at $T = 140$ K after a thermal cycle, as indicated by the labels in **a** (after a quench from $T = 300$ K (**b**), after a slow cool from $T = 420$ K (**c**), after a quench from $T = 300$ K (**d**) and after a quench from $T = 420$ K (**e**)). Red and blue markers highlight the C-CDW and H-CDW-like satellite peaks. Green dashed lines highlight the a^* and b^* directions. **f**, Ag Kα reciprocal space maps parallel to the [*H*, *K*, 0] plane ($\delta L = 0.5$ and $L = 10$) as a function of temperature across the C–NC transition. The black arrows encode the a^* direction. **g**, Temperature dependence of the relative change of the *c*-axis lattice parameter during cooling (blue) and heating (red) for a quenched sample (circular markers). Grey solid lines show the same magnitude for an as-grown sample. The quenched crystal shows a narrow hysteresis window for the C–NC transition compared to an as-grown sample. **h**, Temperature dependence of a H-CDW-like satellite peak ($L = 9.66$ and $\delta L = 0.1$) during cooling (blue markers) and warming (red markers) The thermal hysteresis follows that of the *c* axis. **i**, Temperature dependence of a H-CDW-like-precursor satellite peak ($L = 9.33$ and $\delta L = 0.1$) in the NC phase during cooling (blue markers) and warming (red markers). Error bars in **h** and **i** represent one standard deviation over five different measurements.

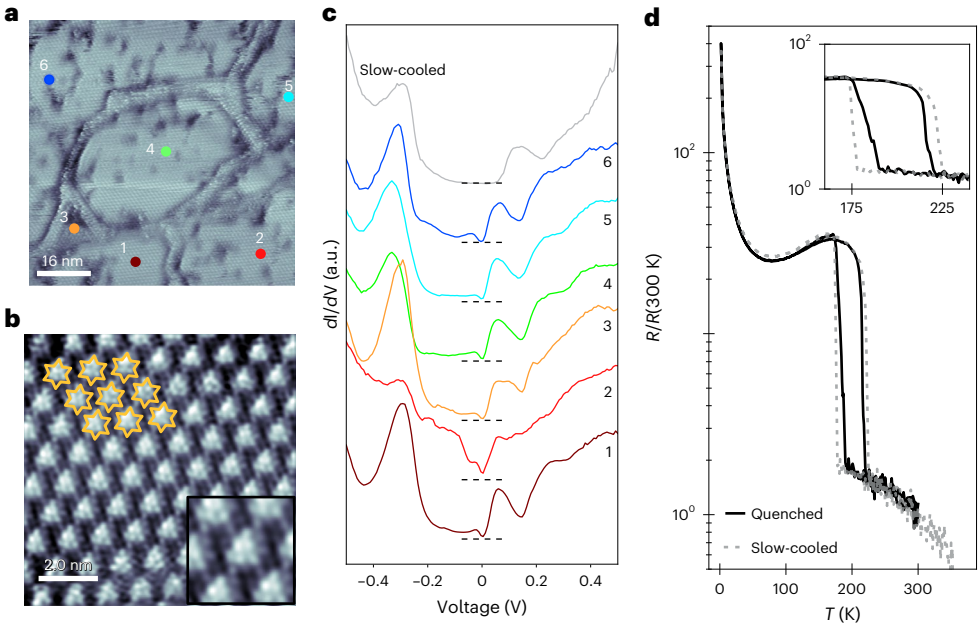

Fig. 3 | Electronic structure of the mixed-CDW phase. a,**b**, Representative topography of the mixed-CDW phase showing domains (**a**), atomic resolution (**a**) and the $\sqrt{13} \times \sqrt{13}$ reconstruction (**b**). **c**, d*I*/d*V* curves for each of the six domains seen in **d**. The d*I*/d*V* data reflect a narrower gap in the electronic structure of 1T-TaS_2 compared to the slow-cooled C-CDW phase (grey). Metallic behaviour was observed in the domain walls, as indicated by point 3. **d**, Resistivity, normalized to that at $T = 300$ K for a mixed-sample (solid line) and an as-grown sample (dotted grey). Inset, zoom-in view around the transition.

between the C-CDW and the H-CDW-like domains. However, from the Fourier transform of STM images with atomic resolution, we extracted a rotated satellite peak $\phi = 12.8(5)°$, which reflects the presence of domains of opposite chirality (related by a mirror operation along one of the high-symmetry crystallographic directions) (Extended Data Fig. 4). Our STM data resemble the in-plane inhomogeneity observed in optically and electrically driven 1T-TaS_2, where several domains of opposite phase have been reported[26,30,31]. Note that the lower domain density prevented us from performing a comprehensive analysis of the in-plane phase configuration of the chiral domains, but we cannot exclude the presence of phase domains (related by a translation operation)[26,30,31].

To summarize, intermediate-quenched bulk 1T-TaS_2 samples host a mixed-CDW state with two distinct order parameters locked to opposite chiral phases, which emerge from a multi-chiral NC-CDW phase. Moreover, our temperature-dependent data revealed the narrowing of the hysteretic window across the NC to mixed-CDW phase transition. This indicates a stronger out-of-plane electronic correlation in the mixed-CDW phase of 1T-TaS_2.

Electronic structure of the bulk quenched samples

We next considered the low-temperature electronic structure of the mixed-CDW phase. Although the origin of the insulating nature of the C-CDW phase remains controversial[47,48], it has been well established that the *c*-axis stacking of CDW domains strongly influences the electronic structure of 1T-TaS_2 (refs. 49–52). Moreover, recent STM and STS studies of the stacking shift between in-plane domains in consecutive layers have identified interlayer coupling as the key ingredient that determines the band structure of the low-temperature CDW phase in 1T-TaS_2 (refs. 26,30,53–55). Thus, we used STS to study the low-temperature electronic state of the quenched samples. dI/dV curves taken at $T = 10$ K within six different domains (Fig. 3a) reveal a reduction of the energy gap $E_g = 300$ meV of the C-CDW[10] towards a state with a reduced but non-zero density of states at the Fermi level (Fig. 3c). Although a transition to a semimetallic state is consistent with that observed for H-CDW[26,30], the density of states maintains two pronounced peaks at around $V_B = -300$ and 150 meV, which have been associated with the lower and upper Hubbard bands[56–58]. Note that the finite in-plane density of states at the Fermi energy E_F and the formation of metallic domain walls are not consistent with a pristine C-CDW phase (Fig. 3b, curve 3) and further supports a mixed-phase state.

On the other hand, the electrical transport in the mixed-CDW sample resembled that in the as-grown crystals. There was insulating behaviour at all temperatures with the same relative resistivity increase with respect to R(300 K) as the C-CDW phase[1] (Fig. 3d), with a narrower thermal hysteresis that tracked that of the structural transition (Fig. 2g) and reflected the role of out-of-plane order in the resistivity of 1T-TaS_2. Although there were metallic domains in the mixed state, the insulating nature of the bulk mixed-CDW phase was distinct from the substantial $T < 70$ K resistivity drop observed for the light-induced H-CDW phase in thin samples[22]. This difference probably originated from the insulating C-CDW domains, which occupied a large volume fraction of the mixed-CDW state, and from stacking CDW disorder between the two CDW orders with different periodicities along L (Fig. 2g,h). Additionally, we found that in situ thermal cycling above $T_{IC} = 350$ K followed by slow cooling recovered the characteristic resistivity of bulk 1T-TaS_2 with a hysteretical transition separated by $\Delta T = 50$ K (Fig. 3d), consistent with our diffraction measurements in as-grown samples. Thus, our STS, resistivity and XRD data support an interpretation in which bulk metallic transport is impeded by the stacking disorder along L of two order parameters with different long-range intralayer CDW orders[59].

Discussion

In summary, based on the measured ordering wavevector and the observed metallic in-plane density of states, our results demonstrate that a thermal quench can stabilize a mixed-CDW phase characterized by C-CDW and H-CDW domains with opposite chiralities through $T_{I\text{-}NC}$. The required thermal quench was at an intermediate rate (~120 K s^{-1}) based on heat diffusion calculations (Methods), faster than the initial quench that 1T-TaS_2 crystals experience during growth (≤2 K s^{-1}) but slower than the changes to the electronic temperature induced by a laser pulse (≥1 K fs^{-1})[6,22] (Fig. 4a). Note that when it was stabilized by ultrafast optical or electric pulses, the H-CDW phase inherited the same chirality as the C-CDW phase. This was probably due to strain[6] or preferential homochiral CDW stacking[16] locally pinning the chirality in the length scale of the laser beam spot or due to the electrode inducing the H-CDW phase. In our quenched samples, the whole sample volume was driven through $T_{I\text{-}NC}$, the mirror-symmetry-breaking transition, thus enabling the spontaneous formation of a multi-domain state. The chiral splitting (relative rotation) of the two order parameters enabled us to clearly resolve the mixed-CDW phase. If the H-CDW and C-CDW phases had exhibited the same chirality, the small difference in their in-plane ordering wavevectors would lead to overlapping satellite peaks, complicating the observation of this new mixed state. Additionally, our STS data reveal that this mixed state hosted a semimetallic in-plane electronic density of states. However, the out-of-plane CDW stacking disorder prevented bulk metallic behaviour.

It is compelling to compare our observations with theories for dynamical phase transitions[60,61] and to assess how domain formation influences the competition between coexisting order parameters. Thus, to describe the low-temperature state after a quench, we considered a mean-field energy functional[62] within the Landau–Ginzburg theory including, among others (see the Supplementary Information for more details of the calculations), a $d_0\bar{\phi}_C^2\bar{\phi}_H^2$ term in which d_0 parameterizes the competition between the C-CDW and the H-CDW phases. The order parameter is the spatial charge modulation, $\alpha(\mathbf{r}) = \Re(\phi_C\, e^{i\mathbf{Q}_C\cdot\mathbf{r}} + \phi_H\, e^{i\mathbf{Q}_H\cdot\mathbf{r}})$, in which the amplitude of the order parameter for the C-CDW, ϕ_C, corresponds to the global minimum of the free-energy landscape for slow-cooling protocols, and ϕ_H is the amplitude of the H-CDW order parameter. Note that our model does not include the order parameter for the NC-CDW as the initial quench from $T_{I\text{-}NC}$ down to low temperature effectively skipped over the intermediate CDW state. Moreover, the H-CDW phase could be stabilized only from the C-CDW phase[22], indicating that it is not necessary to consider NC-CDW when discussing the competition between these two states. This free-energy landscape allows a mixed inhomogeneous phase to be accessed at intermediate but adiabatic quenching rates in which a single temperature can be defined for all the different degrees of freedom in the material. This state is characterized by the coexistence of trapped regions of the two competing orders[63] and enhanced lifetime due to the transiently created domain walls[60].

These behaviours are illustrated in Fig. 4b,c. The energy functional $F(E)$ was calculated for $d_0 = 0.8$ with the same mass term a_0 for both order parameters, $\frac{a_0}{2}\int \mathrm{d}r^2\alpha(\mathbf{r})^2 = \frac{a_0}{2}(\bar{\phi}_C^2 + \bar{\phi}_H^2)$. Under these conditions, the mixed state becomes the global minima ($F(E)_{mixed} = -52.17$), and we found a large energy barrier to the next local minima corresponding to a pure C-CDW state ($F(E)_C = -5.66$). By contrast, a much larger value of $d_0 = 6.1$ led (Fig. 4d,e) to a state in which an almost pure H-CDW phase emerged as a local minimum ($F(E)_H = -2.28$), although with a much shallower energy barrier ($F(E)_C = -5.66$). Note that d_0 encodes the number and size of adjacent domains with different order parameters that emerged during the quench. It is proportional to the length of neighbouring domain walls of the two competing orders. Thus, large values of d_0 are associated with larger populations of small domains, expected in fast quenches. Small values of d_0 indicate the existence of larger domains, consistent with intermediate quenches. Note that for values of $d_0 < 0.5$, competition was not possible, and there was only a global minimum corresponding to the C-CDW phase within our model. The dependence of the global minima on d_0 and a_0 is shown in Fig. 4f, where the colour encodes $F(E)_{mixed}/F(E)_H$.

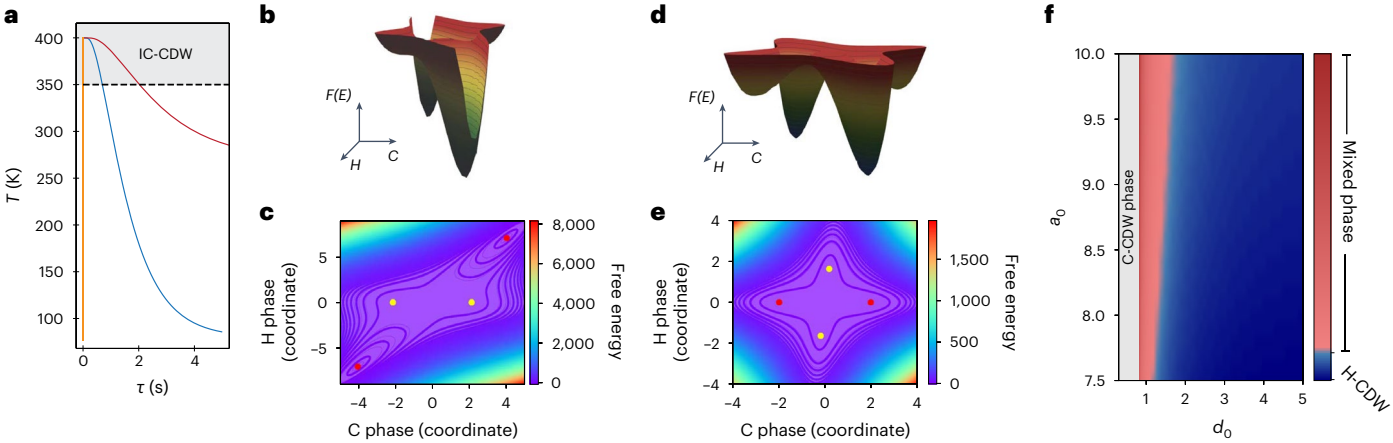

Fig. 4 | Dynamic phase transition with two competing order parameters in 1T-TaS_2. **a**, Cooling rate estimates for an optically induced quench (orange) and solutions of the heat diffusion equation for the slow (red) and fast (blue) cooling schemes used in this work (Methods and Extended Data Fig. 1). **b**, Mean-field-only free-energy functional for the two competing CDW orders, with a competition factor $d_0 = 0.8$ corresponding to an adiabatic quench. **c**, Two-dimensional projection of the energy functional in **b**. **d**, Mean-field-only free-energy functional for the two competing CDW orders, with a competition factor $d_0 = 6.1$ corresponding to a laser quench. Fast quenches correspond to larger values of d_0. **e**, Two-dimensional projection of the energy functional in **d**. The yellow and red markers indicate local and global minima, respectively. **f**, Computed phase diagram as a function of the competition factor d_0 and mass term a_0. The mixed-CDW phase could be stabilized only in a narrow range of intermediate-rate quenches. Faster quenches (optical or electrical pulses) led to the suppression of the C-CDW phase and the emergence of the H-CDW phase.

Our data are consistent with a scenario in 1T-TaS_2 in which a thermal quench stabilizes a mixed phase of H-CDW and C-CDW order parameters up to $T = 180$ K during cooling and $T = 210$ K during warming. Note that coexistence of the H-CDW and C-CDW phase in ultrafast switched samples has also been recently reported[64]. Moreover, all our thermal quenching attempts were unsuccessful in engineering a sample without signatures of the C-CDW state, indicating that sufficiently large values of d_0 were not achieved. It is probable that the quench protocol we describe here resulted in a non-uniform quench rate that was locally too slow over some regions of the sample to allow for the nucleation of the H-CDW domains. Within this context, the previous ultrafast stabilization of the H phase in thin flakes[22–29] could be understood as a non-thermal ultrafast quench[60,61] leading to high domain densities and large values of d_0. Our results are consistent with proposals in which the absorption of energy by the electronic degrees of freedom suppresses the C-CDW order parameter. The consequent relaxation of the non-thermal electrons through the lattice leads to an ionic displacement from that of the C-CDW state into that of the H-CDW[31], allowing for the correlations of the H-CDW order parameter to grow[60]. Thus, our results and Landau–Ginzburg theory calculations indicate that the minimum of the free-energy landscape associated with the H-CDW order parameter already exists at the thermodynamic equilibrium of 1T-TaS_2 and can be accessed with a sufficiently fast adiabatic quench (Fig. 4a).

Our quenching protocol and the observation of the mixed-CDW phase opens a pathway towards the development of new efficient functional devices based on 1T-TaS_2 in which H and C-CDW domain-engineering can lead to switchable metallic behaviour at temperatures above LN_2 (refs. 28,40,41). More generally, our results have implications for the dynamic control of quantum materials[39,65] by establishing that, for macroscopic order parameters, the engineering of free-energy landscapes can be achieved without the need for strong laser pulses when a thermally stable phase is already present at equilibrium[66–68].

Online content

Any methods, additional references, Nature Portfolio reporting summaries, source data, extended data, supplementary information, acknowledgements, peer review information; details of author contributions and competing interests; and statements of data and code availability are available at https://doi.org/10.1038/s41567-025-02938-1.

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Methods

Sample growth

1T-TaS_2 bulk single crystals were grown by chemical vapour transport, with iodine as the transport agent. Stoichiometric amounts of elemental Ta and S were sealed in a quartz tube. The tubes were held at a 950–850 °C temperature gradient for 240 h and then placed in ice water to stabilize the 1T phase. Samples were characterized by XRD, resistivity and Raman spectroscopy (Extended Data Fig. 2).

Thermal quench

Bulk 1T-TaS_2 crystals were sealed in a 1-inch quartz tube filled with Ar gas. The 1T-TaS_2 samples were then annealed by warming them up to 420 K, which was below the polytype transition at 660 K, with an 8-h hold. This was followed by either a fast quench in which the quartz tube was dropped in a reservoir of LN_2 or by slow cooling at room temperature. The in situ quench was performed at the QM2 beamline of the Cornell High Energy Synchrotron Source (CHESS) by alternating between hot gas and a LN_2 cryo-stream.

Estimating the cooling rate

To estimate the cooling rate for our 1T-TaS_2 sample quenched within an Ar-filled quartz ampoule, as shown in Fig. 4a and Extended Data Fig. 1, we solved the heat diffusion equation for a system with cylindrical symmetry under the assumption that 1T-TaS_2 was at all times thermalized with the Ar bath and that the outside surface of the quartz tube was thermalized with the bath. Thus, the limiting factor to thermalization was heat diffusion from the hot Ar volume at the starting temperature ($\rho_{Ar} = 1.8$ kg m^{-3}, $K_{Ar} = 0.0179$ W m^{-1} K^{-1} and $C_p^{Ar} = 400$ J kg^{-1} K^{-1}) through the 1.1-mm-thick quartz wall, with the outer surface acting as the heat sink ($\rho_Q = 2{,}650$ kg m^{-3}, $K_Q = 1$ and $C_p^Q = 700$ J kg^{-1} K^{-1}).

X-ray HDRM

Hard X-ray (20 keV) HDRM was performed at the QM2 beamline of CHESS using a Pilatus 6M area detector. The sample was mounted on a Kapton pin using GE varnish and rotated around three different axes by 360°. Data were collected at every 0.1° step with a frame rate of 0.1 s. The data were reduced into a three-dimensional stack, indexed and projected into a two-dimensional dataset using the beamline software.

In-house XRD measurements

In-house XRD measurements were performed on a Huber four-circle diffractometer in combination with a micro-focused Ag Kα X-ray source and a 256 × 256-pixel GaAs detector (pixel size 55 × 55 μm) situated 96 cm away from the centre of rotation. The data were reduced using the same software as used for the CHESS data.

STM and STS measurements

STM and STS measurements were performed on a closed-cycle scanning tunnelling microscope (RHK Technology PanScan Freedom). Samples for the STM and STS measurements were mounted on stainless steel post holders using conductive silver epoxy and quenched as previously described. After the quench, the samples were characterized by XRD and kept below $T = 370$ K. The surface of a 1T-TaS_2 crystal was cleaved at room temperature and ~10^{-10} mbar by carbon tape and a cleaving screw before being introduced to the STM stage and cooled to 10 K. All data presented here were taken at 10 K.

Resistivity

Resistivity measurements were performed using a physical properties measurement system (Quantum Design). Samples were mounted on a d.c. resistivity puck and a standard four-probe method was used with Au wires and Ag paste contacts.

Raman spectroscopy

Raman measurements of quenched and unquenched 1T-TaS_2 single crystals were performed at 5 K in a closed-cycle He cryostat. The samples were excited in a backscattering geometry with a 532-nm laser focused through a objective lens with a 0.6 numerical aperture and ×40 magnification. The laser power was measured to be 110 μW before the objective. The scattered light was collected through the same lens and directed to an imaging spectrograph with an LN_2-cooled charge-coupled device. Rejection of the Rayleigh scattered light was accomplished using three-volume Bragg gratings that allowed measurement of Raman photons down to 10 cm^{-1}.

Data availability

The X-ray scattering data, STM data and resistivity data generated in this study are available via Zenodo at https://doi.org/10.5281/zenodo.15374771 (ref. 69).

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Author contributions

A.d.l.T and Q.W. synthesized and characterized the single-crystal samples. A.d.l.T., Q.W., K.W.P. and J.P.C.R. performed the high-dynamic-range X-ray scattering measurements. All X-ray experimental data were analysed and interpreted by A.d.l.T. and K.W.P. B.C., J.V.R. and S.M.H. performed and analysed the STM and STS measurements. D.B. and P.M.V. performed the cryo-Raman experiments. Y.M. and G.A.F. performed the Landau–Ginzburg free-energy calculations. A.d.l.T. and. K.W.P. wrote the manuscript with input from all co-authors.

Competing interests

The authors declare no competing interests.

Additional information

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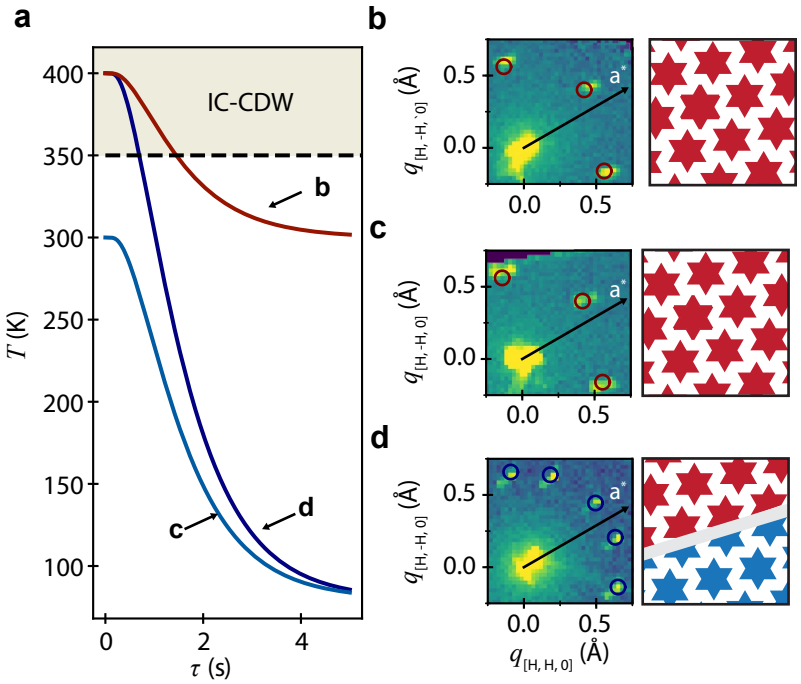
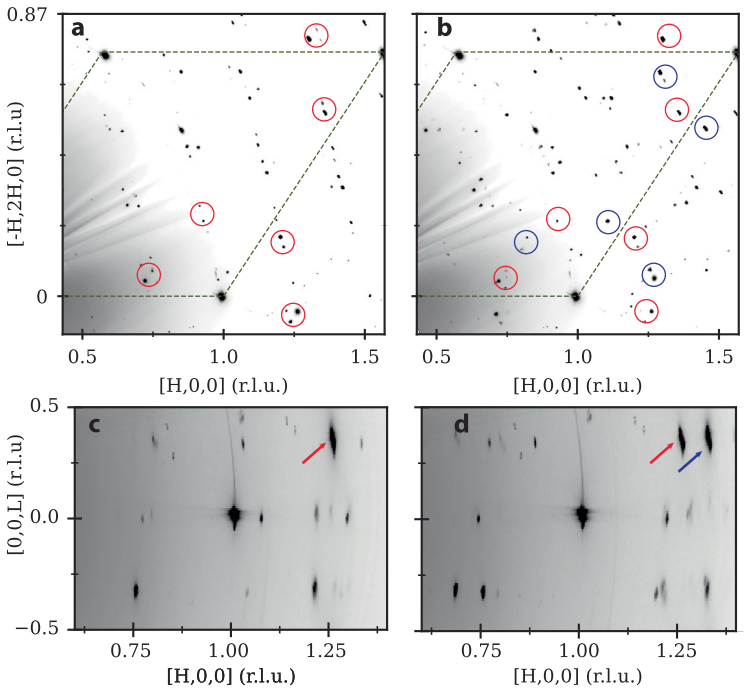


Extended Data Fig. 1 | Cooling rate dependence of the mixed-CDW state. a Cooling rate simulations and reciprocal space maps and CDW state schematic **b**-**d** for three different scenarios: **b** slow cool from 400 K, **c** quench from 300 K, and **d**, quench from 400 K. Only the third case shown in **d** results in the emergence of the mixed CDW state.

https://doi.org/10.1038/s41567-025-02938-1



Extended Data Fig. 2 | Evidence for a two-domain NC-CDW phase. a. Reciprocal space maps parallel to the [HK0] plane (δL = .5, L = 0) at T= 300 K for an as-grown sample. **b**. Reciprocal space maps parallel to the [HK0] plane (δL = .5, L = 0) at T= 300 K after a quench. Red and blue markers highlight the original NC-CDW and the second set of satellite peaks that emerge after the quench, respectively. Green dashed lines highlight the a* and b* directions. **c**. Reciprocal space map parallel to the [H0L] plane for the as-grown sample (δK =.15, K= 0). **d**. Reciprocal space map parallel to the [H0L] plane for the quenched sample (δK =.15, K= 0). Red and blue arrows point to two sets of out-of-plane Bragg peaks corresponding to the NC-CDW. The L dependence and central value of these peaks are different than those seen in the mixed phase sample for the H- CDW phase.

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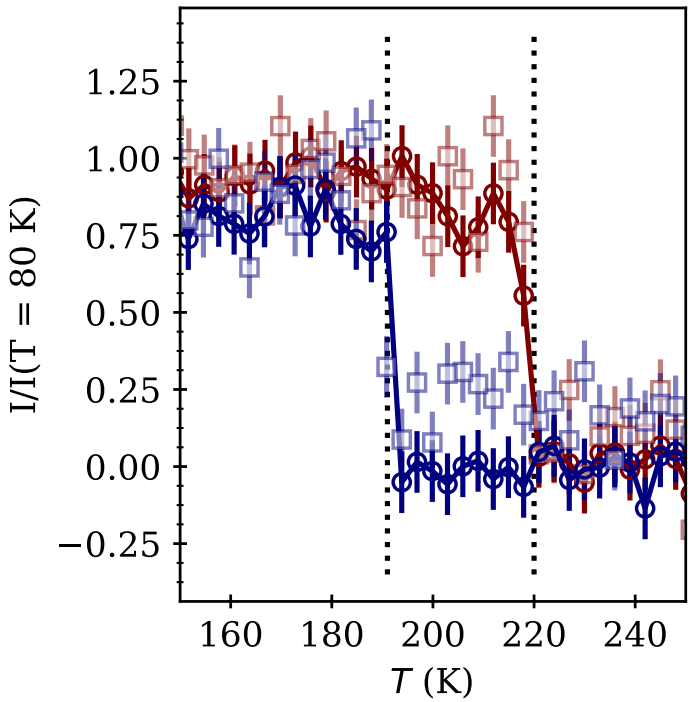

Extended Data Fig. 3 | Temperature dependence of the CDW Bragg peaks. Circular and square markers show the temperature dependence in cooling (blue) and warming (red) of two CDW Bragg peaks of opposite chirality in quenched samples of 1*T*-TaS_2.

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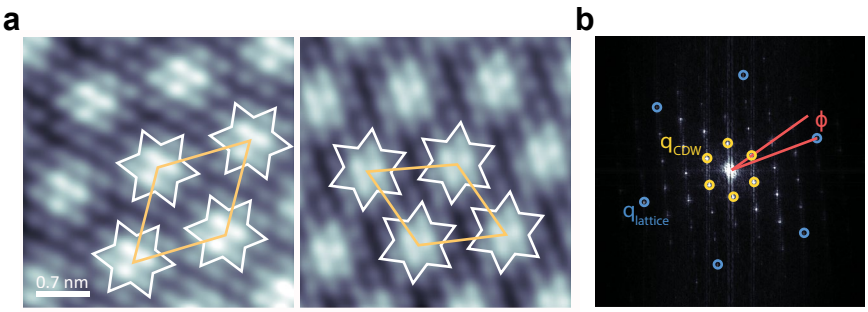

Extended Data Fig. 4 | Chiral domains in STM. a STM data with atomic resolution in two domains with different chirality in Fig. 3b. **b** The high-intensity fundamental peaks ($q_{lattice}$) are highlighted in blue. The satellite peaks (q_{CDW}, yellow) are associated with the mixed-CDW phase.The resolution of our STM measurements prevents us from resolving the difference in the magnitude and in-plane rotation of |q| between the C-CDW and H-CDW.

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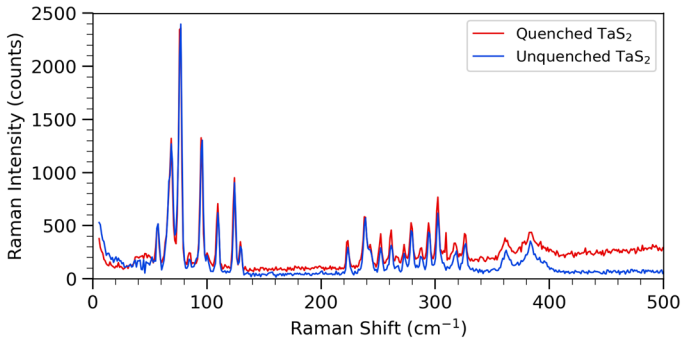

Extended Data Fig. 5 | Raman characterization. Unpolarized Raman spectra in a quenched (red) and as-grown (blue) at 5 K.